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INTERNATIONAL GCSE

Biology

9201/1

Paper 1

Mark Scheme

November 2022

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2 2 B Y 9 2 0 1 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	light		1	AO1 3.1.1b 3.3.1a 3.2.1a, b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	Human Rabbit Lettuce		1	AO2 3.3.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	$\frac{1914}{17\,4000} \times 100$ 1.1 (%)		1 1	AO2 3.3.1a 6.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.										
01.4	<table><thead><tr><th>Cell structure</th><th>Function</th></tr></thead><tbody><tr><td>Mitochondria</td><td>Contains cell sap</td></tr><tr><td>Nucleus</td><td>Controls the activity of the cell</td></tr><tr><td>Vacuole</td><td>Controls what enters and leaves the cell</td></tr><tr><td></td><td>Releases energy for the cell</td></tr></tbody></table>	Cell structure	Function	Mitochondria	Contains cell sap	Nucleus	Controls the activity of the cell	Vacuole	Controls what enters and leaves the cell		Releases energy for the cell	do not accept more than one line from a box on the left	1 1 1	AO1 3.1.1a 3.1.1b
Cell structure	Function													
Mitochondria	Contains cell sap													
Nucleus	Controls the activity of the cell													
Vacuole	Controls what enters and leaves the cell													
	Releases energy for the cell													

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	fewer (trophic) levels / stages	allow the converse argument if clearly referring to Fig 1. allow shorter food chain allow fewer (number of) different organisms	1	AO2 3.3.1d
	(so) less energy / biomass is lost (between stages of the food chain)		1	AO3 3.3.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	decomposition	in this order only	1	AO1 3.3.3a 3.3.3b
	microorganisms	allow bacteria / fungi / decomposers	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	carbon dioxide		1	AO2 3.3.3c 3.3.3e 3.2.1a,g
	nitrate ions		1	

Total			13	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	B		1	AO2 3.1.5a, c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	(aerobic) respiration	allow oxidation	1	AO1 3.1.5c 3.2.6a,c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	A		1	AO2 3.1.5f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	energy		1	AO1 3.1.5f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	10 cm ³ of (1.0 mol/dm ³) sugar solution and 40 cm ³ (distilled) water	allow correct multiples	1	AO4 3.1.5d RPA1
	using 50 cm ³ or 100 cm ³ measuring cylinder or measuring pipettes		1	6.1 6.3.6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	concentration of sugar solution		1	AO4 3.1.5d RPA1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	it does not fit the pattern	allow the value is too low allow it does not fit on / near line of best fit	1	AO4 3.1.5d RPA1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	any one from: • reading of end mass was too low • reading of initial mass was too high • blotting removed too much water from potato piece • sugar concentration was made too high • potato piece not submerged in solution	ignore reference to human error or incorrect reading unqualified	1	AO4 3.1.5d 6.1 RPA1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.9 (view with Figure 4)	suitable curve of best fit which ignores point at 0.2 mol/dm ³		1	AO2 3.1.5d
	concentration inside the potato cells is 0.64 (mol/dm ³)	allow figure from where their line of best fit crosses 0 % change. allow a tolerance of $\pm\frac{1}{2}$ small square	1	AO4 6.1 RPA1

Total			11	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	mitosis		1	AO1 3.5.2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	growth		1	AO1 3.5.2d
	replacement	allow repair of tissues / organs do not accept repair of cells	1	3.5.2j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	reproductive organs	allow testes and ovaries allow gonads	1	AO1 3.5.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	halved		1	AO1 3.5.2h 3.5.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	23 (pairs)	ignore 46	1	AO1 3.5.2c 3.5.3c,d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	(different) forms / versions of a / one gene	do not accept forms of genes	1	AO1 3.5.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	cells that can be specialised / differentiated		1	AO1 3.5.2k
	into many / other types of (human) cell		1	

Question	Answers	Mark	AO/ Spec. Ref
03.8	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO3
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4	AO3
	Level 1: Relevant points are made. They are not logically linked.	1–2	AO2
	No relevant content.	0	
	Indicative content accept converse argument if clearly referring to nerve block injections. Stem cell treatment: • stem cells are taken from the patient's bone marrow • (so) no possibility of the body rejecting the cells or the new tissue formed • (and) no risk of allergic reactions • stem cells encourage damaged cells to repair themselves • (so) repair would be permanent • success rate is higher than for nerve block injections • (so) more patients are relieved of back pain • only one procedure is needed • (so) reduced discomfort over time • less risk / fewer side effects associated with the procedure • (so) less chance of patient needing additional treatment • x-rays are carcinogenic • (so) can cause tumours / cancer • (repair) process can take many weeks to occur • (so) patient will be in pain for longer (than nerve block injections) immediately after treatment • newer procedure • (so) less time to see (possible) long term effects		3.5.2l 3.5.2m 3.5.2n 3.5.2o
Total		15	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	structure	in either order	1	AO1 3.1.2b
	function	allow job	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	tissues		1	AO1 3.1.2c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	an organ system		1	AO1 3.1.2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	3:1		1	AO2 3.1.5h 6.3(6)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	organism B		1	AO2 3.1.5h 6.3(6)

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	to keep the blood flowing in the correct / one direction	allow to prevent the blood flowing backwards	1	AO1 3.2.3h
		allow to prevent backflow of blood		
		allow to keep blood flowing forwards		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.7	the valve tissue may stiffen preventing the valve from closing properly	allow the valve may stiffen causing backflow of blood	1	AO1 3.2.3f
	the valve may develop a leak allowing blood to mix	allow the valve may develop a leak allowing blood to flow backwards if no other mark awarded allow valves may become stiff and valves may leak for 1 mark	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.8	(dead) humans / animal tissue metal / plastic	this order only	1	AO2 3.2.3f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.9	artificial heart	allow ventilator if linked to heart not having to work so hard allow LVAD	1	AO1 3.2.3g

Total			11	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	sound	this order only	1	AO2 3.4.6e
	visual		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	innate		1	AO2 3.4.6c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	any three from: • laying up to 25 eggs in one season means many offspring or • large number of eggs in one season means many offspring • laying eggs which look like host species eggs means host looks after eggs • distracting host female and / or laying eggs very quickly means host bird does not notice the new egg • laying eggs which hatch early means chicks get more food • having chicks with loud begging call means chicks get more food • having chicks which grow quickly means they are independent earlier		3	AO2 3.4.6b 3.6.2b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	do not waste time / energy in finding new mates		1	AO3 3.4.6a,b 3.6.2b
	(so) eggs can be laid earlier in season or (so) more eggs can be laid		1	
	(so) earlier chicks get more of the food supply or more offspring are hatched each year		1	
Total			9	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	LHS $C_6H_{12}O_6$ $6O_2$	in any order	1	AO1 3.2.6b 3.2.6c
	RHS $6CO_2$ $6H_2O$	in any order allow 1 mark if all chemical formulae are correct but not balanced	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	as running speed increases, rate of oxygen absorption increases		1	AO2 3.2.6h
	then levels off / plateaus	allow becomes constant	1	
	at 13 km per hour or at 9.6 cm ³ per minute	allow value in range 13 to 14 km per hour allow other use of comparative data	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	(heart beats faster for)	reference to 'increased' must be made at least once for full marks		AO1 3.2.6a 3.2.6b 3.2.6g 3.2.6h
	increased blood flow		1	
	(to bring) more oxygen	allow (to bring) more glucose	1	
	(for) increased respiration		1	
	(for) increased release of energy for muscle contraction	do not accept produces / makes / creates energy	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	not enough oxygen reaching the muscles	allow cannot absorb enough oxygen for aerobic respiration	1	AO3
	(so) anaerobic respiration occurs		1	AO1
	(which produces) lactic acid		1	AO1 3.2.6a 3.2.6i 3.2.6j 3.2.6l

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	supply the oxygen needed to pay off the oxygen debt		1	AO2 3.2.6j 3.2.6l
	(so) lactic acid is broken down / oxidised to carbon dioxide and water		1	
		if neither mark awarded allow to supply oxygen to break down / oxidise lactic acid for 1 mark		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.6	carbon dioxide and ethanol	allow formulae ignore alcohol	1	AO1 3.2.6m

Total			15	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	pathogens		1	AO1 3.4.7a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	viruses live in cells so it is hard for drugs to access them or drugs that kill viruses will also damage cells / tissues (of the body)		1	AO3 3.4.7b 3.4.7g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	bacteria mutate (quickly)		1	AO1 3.4.7h
	which produces some (strains of) bacteria resistant to the antibiotics being used		1	AO1 3.4.7h
	so these (strains) survive, reproduce and spread, needing a new antibiotic to treat the infection		1	AO3 3.4.7j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4 View with Table 2	conversion of equation to $\text{distance} = \frac{1}{\text{concentration}}$		1	AO2 3.4.7j 6.3.17
	$\frac{1}{4.0}$		1	6.3.10
	0.25 (cm)		1	6.3.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	inverse (proportion / correlation)		1	AO3 3.4.7j 6.3.18

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6 View with Figure 10	area of a circle is $A = \pi r^2$		1	AO2 3.4.7j RPA5 6.3.14 6.3.12
	from graph or by using equation $r = 0.67$	if graph used allow tolerance of $\pm \frac{1}{2}$ small square	1	
	$= 3.14 \times 0.67 \times 0.67$		1	6.3.10
	$= 1.4095.... \text{ cm}^2$		1	6.3.3
	$= 1.41$	allow their answer correct to 3 sig figs if a reasonable attempt has been made at calculation. an error in one step does not prevent award of marks in subsequent steps	1	6.3.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.7	10% of the bacteria have been killed	allow it shows 200 out of 2000 have been killed	1	AO2 3.4.7 6.3.19

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.8	vertical line at 0.9 cm (going at least as high as 1200)		1	AO2 3.4.7 6.3.19

Total			16	
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